

Beckmann, M., Beinert, R., and Bresch, J. (2025) – Max-normalized Radon CDT for limited data classification, **Lect. Notes Comput. Sci.**

Beckmann, M., Beinert, R., and Bresch, J. (2025) – Normalized Radon CDT for invariance and robustness in optimal transport based image classification, **SIAM J. Imaging Sci.**

Beckmann, M., Beinert, R., and Bresch, J. (2025) – Generalization of the normalized Radon CDT for limited data recognition, **arXiv.2512.08099**



Generalization of Normalized Radon Cumulative Distribution Transforms in the Light as Feature Extractors for Image Classification

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Motivation: Classification, Recognition

- Classification with **many classes** and **few samples**
- Assumption: (approx.) **affine** / structured class families
- Need: a) **invariance** to affine transformations
- b) **stability** to deformations
- Application: historic watermarks, automated pattern recognition



Normalization: NR-CDT and Sorting

- For $\mu \in \mathcal{P}_c(\mathbb{X}) \subset \mathcal{P}_2(\mathbb{X})$, R-CDT $\widehat{\mathcal{R}}_{\phi, \theta}[\mu]$ is bounded $\equiv \mathcal{R}_{\phi, \theta}[\mu]$
- **Normalize per slice:**

$$\mathcal{N}_{\phi, \theta}[\mu](t) = \frac{\widehat{\mathcal{R}}_{\phi, \theta}[\mu](t) - \text{mean}(\widehat{\mathcal{R}}_{\phi, \theta}[\mu])}{\text{std}(\widehat{\mathcal{R}}_{\phi, \theta}[\mu])}$$
- **Sorting** via $h : L^\infty(\Theta) \rightarrow \mathbb{R}$ (boundedness-preserving)
- **Define** ${}_h\text{NR-CDT}$: $h(\mathcal{N}_{\phi}[\mu](t, \cdot))$
- On $\mathcal{P}_{\phi, c}^+(\mathbb{X})$ ($\text{std}(\widehat{\mathcal{R}}_{\phi, \theta}[\mu]) \geq c > 0$)

Radon on measures (compact form)

- Restricted Radon transform

$$\mathcal{R}_{\phi, \theta} : \mathcal{M}(\mathbb{X}) \rightarrow \mathcal{M}(\mathbb{R}), \mu \mapsto (S_{\phi, \theta})_{\#} \mu, S_{\phi, \theta} := \phi(\cdot, \theta)$$
- Radon transform, gluing $\gamma \in \mathcal{P}(\Theta)$

$$\mathcal{R}_{\phi, \gamma}[\mu] = \mathcal{I}_{\#}[\mu \times \gamma] \in \mathcal{M}(\mathbb{R} \times \Theta), \mathcal{I}(\mathbf{x}, \theta) := (S_{\theta}(\mathbf{x}), \theta)$$
- Disintegration / adjoint hold

$$\langle \mathcal{R}_{\phi, \gamma}[\mu], g \rangle = \langle \mu, \mathcal{R}_{\phi, \gamma}^*[g] \rangle$$

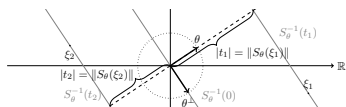
Theory (key statements)

- **Sliced-Wasserstein identity:**

$$\|\widehat{\mathcal{R}}_{\phi}[\mu] - \widehat{\mathcal{R}}_{\phi}[\nu]\|_{\rho \times \gamma}^2 = \int_{\Theta} W_2^2(\mathcal{R}_{\phi, \theta}[\mu], \mathcal{R}_{\phi, \theta}[\nu]) d\gamma(\theta)$$
- **Translation:** for suitable $(\mathbb{X}, \Theta, \phi)$ specific T measurable on $\mathbb{X}$ correspond to $\hat{T}$ on $\mathbb{R} \times \Theta$:

$$\mathcal{R}_{\phi}[\widehat{T}_{\#} \mu] = \widehat{T}_{\#} \widehat{\mathcal{R}}_{\phi}[\mu]$$
- **Separability theorem (concise):**
 - Invariance** $\mathcal{N}_{h, \phi}[T_{\#} \mu] = \mathcal{N}_{h, \phi}[\mu]$
 - Singleton sets** in ${}_h\text{NR-CDT}$ -space

$$\mathbb{F}_{\mu} = \{T_{\#} \mu\} \xrightarrow{{}_h\text{NR-CDT}} \mathcal{N}_{h, \phi}[\mathbb{F}_{\mu}] = \mathcal{N}_{h, \phi}[\mu]$$
 - if $\mathcal{N}_{h, \phi}[\mu_i] \in L_{\rho}^{\infty}$ differ at least once $\rightarrow$ **separable**
 - Robustness** of ${}_m\text{NR-CDT}$ ($h = \sup_{\theta \in \Theta, \text{a.o.}}$) in Wasserstein.



Core Idea & Pipeline

- Slice with generalized Radon on $\mathbb{X}$:

$$\mathcal{R}_{\phi, \theta}[\mu] = (S_{\phi, \theta})_{\#} \mu \in \mathcal{P}(\mathbb{R}), \mu \in \mathcal{P}(\mathbb{X})$$
- 1D embedding via CDT, wrt. ref

$$\hat{\mu} = \arg \min_{T_{\#} \rho = \mu} \int_{\mathbb{R}} c(s - T(s)) d\rho(s), \text{ atomless } \rho, \mu \in \mathcal{P}(\mathbb{R})$$
- R-CDT: collect sliced CDTs

$$\widehat{\mathcal{R}}_{\phi, \theta}[\mu] \in L_{\rho}^2(\mathbb{R}) \text{ for } \mu \in \mathcal{P}_2(\mathbb{X})$$

